Economy Simulator

Mathematical Models and Algorithms

Alex Yang

November 1, 2026

Abstract

Contents

1 Introduction **2**
1.1 Project Overview 2
1.2 Mathematical Objectives 2

2 Crafting and Production Model **2**
2.1 Mathematical Definitions 2
2.2 Construction of the Consumption Matrix 2
2.3 Construction of the Production Matrix 2

1 Introduction

1.1 Project Overview

1.2 Mathematical Objectives

2 Crafting and Production Model

2.1 Mathematical Definitions

2.2 Construction of the Consumption Matrix

The purpose of the consumption matrix C is to represent the quantities of items required to execute each crafting recipe.

Let n denote the total number of items supported by the game, and let m denote the total number of crafting recipes. The consumption matrix is therefore defined as $C \in \mathbb{Z}_{\geq 0}^{n \times m}$.

Each row i, where $1 \leq i \leq n$, represents an item, while each column j, where $1 \leq j \leq m$, represents a crafting recipe.

Thus, the entry C_{ij} specifies the quantity of the ith item consumed when executing the jth recipe once. If the recipe does not require that item, then $C_{ij} = 0$.

2.3 Construction of the Production Matrix

Similar to the consumption matrix, the production matrix B's purpose is to represent the quantities of items produced after executing each crafting recipe. B has the same dimensions as C, because the rows and columns represent the same items and recipes as C, except the entries may be different because B catalogues the number of item outputs whereas C catalogues the number of item inputs.

Therefore, $B \in \mathbb{Z}_{\geq 0}^{n \times m}$, where the entry B_{ij} specifies the quantity of the ith item produced when executing the jth recipe once. If the recipe does not produce that item, then $B_{ij} = 0$.